

Quiz 4 – Applied OS: Cyber-Defense & Agentic AI

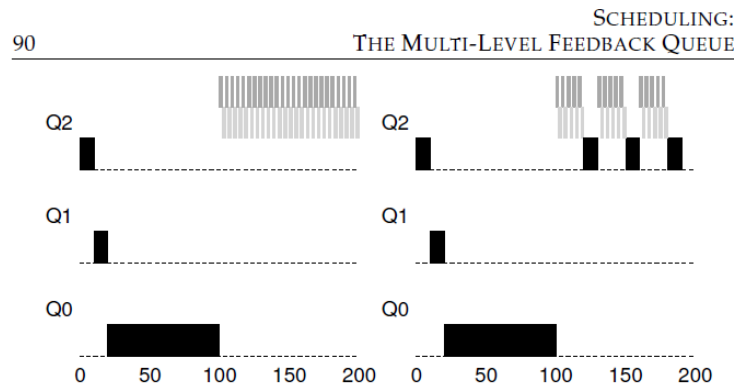
L1: Student Name: _____

Q1			Total
(2)	(5)	(3)	(10)

Q1 What is wrong with this diagram, which shows how MLFQ runs?

a)

(2)



Say Process A is this black process: XXXXXXXXXX

- Now, as seen in the figure above, all processes go back to Q_2 (highest priority queue) after 100 time stamps, so, as per Rule 5 of MLFQ, $S = 100$.
- Once Process A goes to Q_2 it should run for 1 time slice, and then be transferred to Q_1 , and another time slice later, it should be transferred to Q_0 .
- This is NOT being shown in the diagram above.

b) What are the 5 Rules of MLFQ?

(5)

The set of MLFQ rules:

1. **Rule 1:** If $\text{Priority}(A) > \text{Priority}(B)$, A runs (B doesn't).
2. **Rule 2:** If $\text{Priority}(A) = \text{Priority}(B)$, A & B run in RR.
3. **Rule 3:** When a job enters the system, it is placed at the highest priority.
4. **Rule 4:** Once a job uses up its time allotment at a given level (regardless of how many times it has given up the CPU), its priority is reduced (i.e., it moves down the queue).
5. **Rule 5:** After some time period S , move all the jobs in the system to the topmost queue

(c) Write down the code for a basic process structure.

(3)

```
struct proc {  
    int pid;  
    int state;  
    void * program_counter;  
    struct proc * parent;  
    void * base;  
    void * bound;  
    char name[20];  
}
```